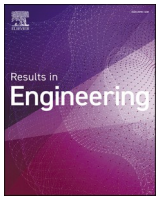
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TinyML-based intrusion detection systems for sustainable and energy-constrained IoT devices

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ABSTRACT

With the exponential growth of the Internet of Things (IoT), ensuring real-time security in energy-constrained and environmentally sensitive environments has become a significant challenge. Traditional Intrusion Detection Systems (IDS), though effective, are often resource-intensive and unsuitable for deployment on low-power edge devices. This research explores the potential of Tiny Machine Learning (TinyML) to provide lightweight, on-device intelligence for threat detection. Research proposes a Gradient Boosting Based Adaptive Artificial Gorilla Troops (GraBoost-AAGT), which combines Adaptive Artificial Gorilla Troops Optimization (AAGTO) and Extreme Gradient Boosting (XGBoost), optimized for ultra-low power consumption and memory efficiency. Preprocessing involves categorical encoding and Z-score normalization, followed by Fast Fourier Transform (FFT) to extract relevant frequency-domain features. Specifically designed for embedded IoT systems, this model, referred to as GraBoost-AAGT, supports real-time detection of cyber threats, such as denial-of-service and probing attacks, while maintaining minimal central processing unit (CPU) and energy overhead. To evaluate performance, network traffic data encompassing both normal and malicious activities is collected using IoT intrusion detection dataset, traffic simulators, and replay tools under controlled attack and non-attack scenarios. Results demonstrate that GraBoost-AAGT deployed via TinyML achieves a high detection accuracy (99.50 %), precision (99.8 %), recall (99.12 %), F1-score (99.45 %), specificity (99.81 %), and computation time (4.5 s), reduced energy consumption, and lower CPU usage compared to traditional edge-based ML models. These findings confirm that TinyML offers a viable and sustainable solution for deploying intelligent threat detection in smart and green IoT ecosystems. This research provides a foundational step toward secure, scalable, and environmentally responsible IoT security frameworks.

Introduction

The growth of highly sophisticated intrusion threats and the rapid expansion of cyber traffic make adaptive and real-time intrusion detection systems essential. Numerous cyberattacks, such as phishing attempts, malware infections, and illegal access to individual information, might result in negative outcomes like compromising private information, financial loss, and damage to one's reputation [1]. Protecting individuals, organizations, and sectors against these classes of cyber-threats requires constant vigilance and aggressive actions. An adaptable and real-time intrusion detection system (IDS) helps businesses stop cyberattacks before they have a chance to cause harm and mitigate the impact of successful breaches by continually monitoring, detecting, and responding to emerging threats [2].

In terms of sustainability, effectiveness, and optimization, the convergence of energy systems and the Internet of Things (IoT) yields important advantages. However, a complex environment of cybersecurity dangers is also introduced by this interconnectedness. The techniques and tools used to recognize and address malicious activity directed at IoT devices used in the energy industry, such as smart grids and renewable energy systems, are referred to as threat detection in this context [3]. Detecting threats and attacks that could harm information networks is becoming more and more important due to the ongoing increase in internet users and the growth of the IoT. Intrusion detection systems are a common intervention used in the field of cybersecurity. A combination of hardware and software technologies for identifying

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intrusions or attacks in computer systems is known as an intrusion detection system [4]. The IoT has advanced as a crucial technology in the digital revolution, enabling effective communication between people and things by connecting smartphones and other small devices to the internet. Wearable technologies may transmit data because they have embedded systems, a variety of sensors, and the ability to connect to a network [5].

It involves systems and methods for cybersecurity that are built to detect and react to criminal activity while using incredibly little energy. For gadgets with constrained power supplies, like battery-powered or energy-harvesting IoT devices, the strategy is crucial. Ultra-low power threat detection offers continuous, real-time security monitoring without considerably using energy by utilizing lightweight models, efficient hardware, and optimized algorithms. This allows for dependable and sustainable protection in areas with limited resources [6]. The lack of integrated privacy and security protections in most digital transformation technologies puts the confidentiality, reliability, and integrity of IoT networks in danger. Because they collect a lot of valuable data, IoT devices and gadgets are more desirable targets for hackers trying to exploit security holes. More vulnerable to malware attacks are IoT devices and gadgets due to their continuous network connectivity. Manufacturers of IoT devices either ignore security and privacy issues or treat them as an afterthought [7].

As a result, creating a precise intrusion detection system is always necessary to effectively counteract various kinds of attacks. In addition to ensuring security and privacy against unwanted access, another major obstacle is that transportation IoT networks are time-efficient, cognitively taxing, and continuously require processing resources [8]. Threat detection for ultra-low power reduces energy usage, which is important for IoT devices that run on batteries or use energy harvesting. This results in a much longer battery life and less need for frequent charging or battery replacements, which saves money and extends the devices' useful lives [9]. While behavior-based and intrusion-based techniques could need more processing power than ultra-low power devices can provide, traditional techniques like signature-based detection have trouble keeping up with changing threats. Energy limitations restrict the frequency and complexity of data analysis that can be carried out, which might make it more difficult to identify threats that are subtle or constantly changing. Although it can be computationally demanding and prone to false alarms, it is effective in detecting hazards that remain to be detected [10]. The research aims to develop a lightweight, ultra-low power Intrusion Detection System (IDS) using Tiny Machine Learning (TinyML) by proposing the Gradient Boosting Based Adaptive Artificial Gorilla Troops (GraBoost-AAGT) method. It targets real-time threat detection in energy-constrained IoT environments, ensuring high accuracy, low energy consumption, and efficient CPU usage, thus enabling secure and sustainable smart IoT systems, especially in green and resource-limited deployments.

Contribution of the research

- To enable ultra-low power consumption and faster convergence by integrating the proposed GraBoost-AAGT method prediction power with metaheuristic optimization for efficiently guiding solution search in complex optimization problems.
- To utilize Extreme Gradient Boosting (GraBoost) for refining candidate solution evaluations, reducing redundant computations, and enhancing learning-assisted optimization with improved accuracy and reduced computational overhead.
- To incorporate Adaptive Artificial Gorilla Troops (AAGT) optimization algorithm for dynamically balancing exploration and exploitation, minimizing memory usage by avoiding population bloating, and maintaining search diversity through adaptive control mechanisms.
- To achieve superior results in constrained environments by combining GraBoost and AAGT, ensuring memory efficiency, faster convergence, and optimal solution quality in ultra-low power applications.

The rest of the research is structured as follows: Relevant research on ML-based intrusion detection for IoT devices is compiled in portion 2. The suggested threat detection methodology is presented in portion 3. Experimental data showing the efficiency of the suggested approach are presented in portion 4. Conclusions as well as suggestions for further research are included in portion 5.

Related work

A Deep Reinforcement Learning (DRL) method to identify malevolent assaults in unmanned aerial vehicle (UAV) aerial computer networks to safeguard the security of UAV networks was proposed by [11]. The context for UAV aerial computing systems and probable usages was presented initially. Next, the topic of intrusion threats in UAV aerial computer networks was covered. It then provided case research of intrusion detection using DRL to safeguard the security services, offering the conclusion along with a number of interesting research avenues.

An efficient and cost-effective audio categorization mechanism based on Convolutional Neural Networks (CNN) [12] was created especially for wireless edge technology with limited resources to lower resource utilization and energy usage for the extensive and efficient identification of unauthorized tree felling. Furthermore, testing with experimental extended transmission was carried out in actual settings. According to data gathered from the testing results, the CNN attained an 85 % accuracy rate in detecting and reporting tree-cutting incidents, enabling cost-effective and productive forest surveillance via smart IoT systems.

The research involved evaluating the performance of several widely used Machine Learning (ML) classifiers on the KDD'99 dataset, including Decision Trees (DT), Support Vector Machines (SVM), Artificial Neural Networks (ANN), Gaussian Naïve Bayes (NB), and Random Forests (RF) [13]. To discuss critical IoT security research issues and explore methods to prevent attacks on IoT platforms, an efficient ML-based IDS aimed at protecting green IoT networks was employed. Among these, the relatively accurate SVM model showed a decline in performance to 90.02 % accuracy, while ANN achieved the lowest accuracy at 85.92 %, indicating the need for more adaptive and lightweight models for sustainable IoT security.

In the context of a reliable and cost-effective IDS for the Internet of Medical Things (IoMT) setting [14], the research demonstrated how a deep recurrent neural network (DRNN) and supervised ML (SML) models could be used to classify and forecast unexpected cyber threats. With an accuracy of 99.76 %, it was determined via extensive evaluation that the suggested SML model works better than previous techniques.

DRL-based IDS that initially leveraged the analytical aspects of previous internet activity to estimate potential traffic tendencies was suggested by [15]. Based on these predictions, the system performed intrusion detection by identifying deviations that could indicate malicious activities. Results from experiments showed how effective the technique was in precisely identifying DDoS attacks, underscoring its potential for protecting networked IoT systems.

An IDS based on DL architecture was used to safeguard the Controller Area Network (CAN) bus in vehicles [16]. The approach utilized the Visual Geometry Group (VGG) architecture, which was trained to recognize various network intrusion patterns for effective detection of malicious attacks. Experiments were conducted with the CAN-intrusion-dataset, and the results demonstrated that the proposed DL-based approach, compared to typical ML approaches, considerably lowered the false positive rate (FPR). The structure attained an overall accuracy of 96 % and an FPR of just 0.6 %.

To provide valuable insights for IoT security, real network environments were simulated to proactively detect and mitigate vulnerabilities [17]. The research focused on identifying anomalies in IoT systems by

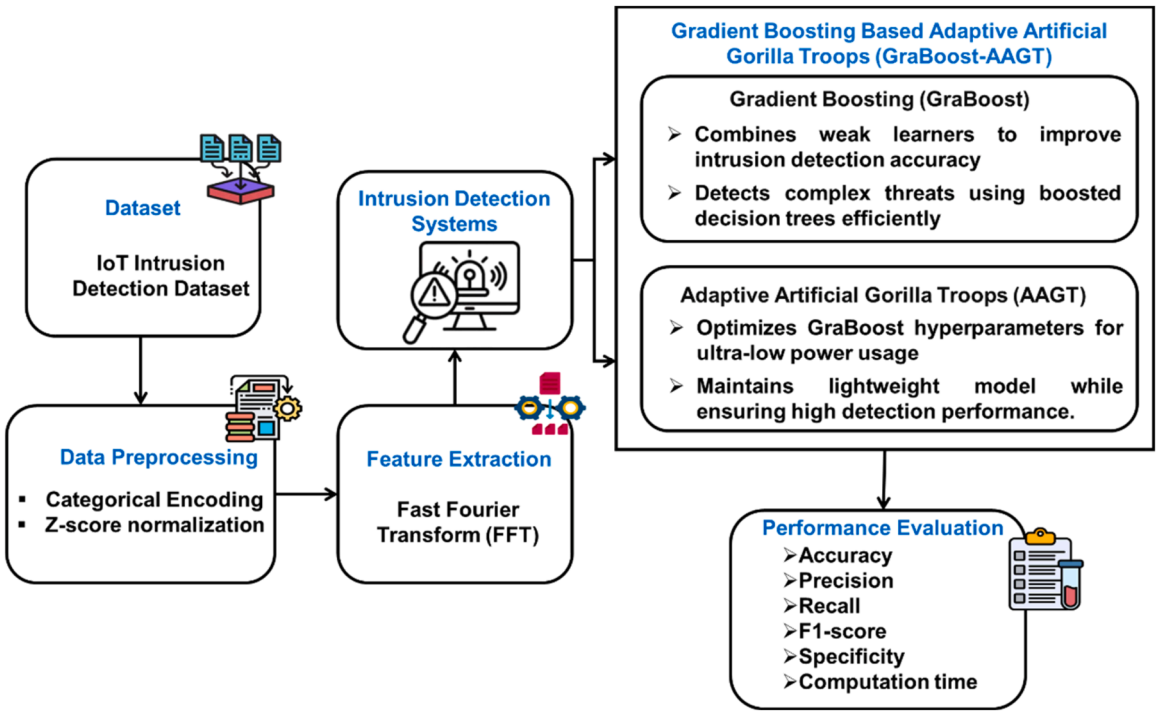


Fig. 1. Framework of the proposed method.

emulating Message Queuing Telemetry Transport (MQTT) protocols over virtual networks. Diverse ML classifiers, including SVM, KNN, and RF, were employed to detect DDoS attacks, with RF achieving the highest performance, reaching 99.94 % accuracy, demonstrating its effectiveness in IoT threat detection.

A hybrid DL technique for Cyber Threat Intelligence (CTI) enhanced threat classification accuracy. The model combined a CNN with a Quasi-Recurrent Neural Network (QRNN), enabling both improved performance and real-time classification [18]. The incorporation of QRNN significantly increased detection speed and accuracy. The research highlighted the model's suitability for deployment in real-time smart system networks, facilitating timely threat mitigation and enhancing overall cybersecurity effectiveness.

A DL-based integrated classification approach to recognize malware in IoT equipment, addressing the ongoing challenge of achieving high detection accuracy. For efficient malware detection, an indicator that combines CNN and Long Short-Term Memory (LSTM) network outputs is used after feature selection and one-hot encoding [19]. Research findings revealed that the proposed approach surpasses existing cutting-edge algorithms on conventional datasets, with an average accuracy of 99.5 %.

The effect of charging strategies on the effectiveness of energy attacks in green IoT networks was examined, and an innovative security issue arising from the integration of renewable energy was identified. Two reinforcement learning (RL)-based techniques, such as Policy Gradient Enabled Intelligent Energy Attack (PG-IEA) and Q-Learning Enabled Intelligent Energy Attack (Q-IEA), were proposed by [20]. Experimental results demonstrated that, compared to networks without attacks, 53.3 % more data traffic was diverted to the deficient network, over four times as much as that of a random attack.

The TinyML uprising undertook an extensive assessment of TinyML-related works, focusing on the usage of both ML and DL models on limited-resource processors [21]. The main contribution of their work lies in the examination of the types of ML techniques employed in TinyML research, alongside detailed insights into the datasets employed, and the types and features of hardware devices.

A method integrating graph-based ML with lightweight cryptography has been proposed to address security and authentication challenges in Intelligent Transportation Systems (ITS) [22]. The suggested approach significantly advanced the creation of an intelligent transportation communication environment by identifying different cyberthreats and authenticating smart vehicles in ITS.

The State-Action-Reward-State-Action (SARSA) Markov Decision Process (MDP) for honeypot strategy was proposed [23]. The suggested system could outperform conventional IDSs in terms of accuracy and convergence speed by utilizing environmental tests. As demonstrated by the data, the suggested model outperformed conventional non-linear IDS, demonstrating substantial accuracy (<0.99) and F-measure (0.98).

To address these issues, an explainable Artificial Intelligence (XAI) Ensemble Transfer Learning (TL) model designed especially for Internet of Vehicles (IoV) zero-day attack detection was suggested [24]. The framework increased computational effectiveness by lowering the density of IoV network data using composite bidirectional long-short-term memory with autoencoders (BiLAE). For real-time applications, the system was very flexible and scalable, and it was made to perform effectively across IoV gateway electronic controllers with minimal computing capability. To solve these problems, IDS that aid in IoT network security could be developed using AI techniques like ML and deep learning (DL) was proposed [25]. On all nine IoT devices, tests of autoencoders (AE) models trained by Federated learning (FL) and those not taught by FL reveal that the FL-trained unsupervised AE model performed the greatest overall across all measures.

Research gap

Despite extensive research on intrusion detection for IoT environments, existing methods exhibit several limitations, including high computational complexity, lack of real-time adaptability, and inefficiency on ultra-low-power or resource-constrained devices. Many studies rely on traditional ML or DL models that require extensive memory, training time, and energy unsuitable for edge-based deployments like TinyML. For instance, while DL models such as CNNs [12], LSTMs [18], or QRNNs [19] provide high accuracy, these methods demand substantial resources. Reinforcement learning (RL) [20]-based

approaches improve adaptability but lack lightweight optimization. To bridge these gaps, the proposed AAGT-GraBoost method integrates AAGT for optimal feature selection with GraBoost for lightweight, adaptive, and high-accuracy intrusion detection. This hybrid approach ensures low power consumption, real-time performance, and efficient deployment in energy-constrained green IoT networks, outperforming existing heavy methods.

TinyML integration shifts intrusion detection from **cloud-dependent to on-device intelligence**, reducing **computational complexity and energy use** while retaining **competitive accuracy**, thereby redefining the efficiency–performance balance in constrained IoT systems

Materials and methods

The suggested method uses the ultra-low power and efficient memory GraBoost-AAGT method, which merges the AAGT optimization algorithm with GraBoost ML based method for precise IoT device intrusive detection. The research utilized publicly available IoT intrusive detection datasets containing both normal and malicious traffic. Preprocessing includes categorical encoding to convert non-numeric categorical features and Z-score normalization, which standardizes the data. Feature extraction is performed using Fast Fourier Transform (FFT), which allows for efficient frequency-domain analysis, which is crucial for detecting anomalous patterns while handling all of the data with minimal computational cost. These processes are displayed in Fig. 1.

Dataset

The IoT intrusion detection dataset includes 1300 IoT network traffic data points specifically designed for intrusion detection systems based on Arduino boards from open source in Kaggle (https://www.kaggle.com/datasets/programmer3/iot-intrusion-detection-dataset). Along with a target label designating traffic as Normal, DoS, or Probe, it contains information on network protocols, packet counts, sizes, and flow duration. The data was split into 80 % training and 20 % testing.

Preprocessing categorial and z-score normalization

Encoding categories involves transforming non-numeric categorical variables into a numerical representation to ensure compatibility with model input layers. In this process, label encoding is employed, wherein each unique category within a feature is mapped to an integer value, preserving the distinctiveness of categorical classes without imposing any ordinal relationship. The alteration is applied across all categorical columns identified by their data types, ensuring the dataset maintains its semantic integrity while becoming fully numeric and suitable for training ML models to detect anomalies across heterogeneous data types.

When characteristics have disparate scales or units, normalization is crucial. For signal data, Z-score normalization, sometimes referred to as standardization, gives each characteristic a mean of 0 and a standard deviation (SD) of 1 for intrusion detection in IoT devices. It is especially important for Eq. (1) and ML models, along with other algorithms that depend on the magnitude of input threat attributes.

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

Where μ indicates the feature's mean, x indicates the initial value of the characteristic, σ demonstrates the SD of the feature, and z is a normalized value.

Feature extraction using fast Fourier transform (FFT)

In the intrusion detection process, significant features are extracted using the FFT for the process of converting time-domain network traffic signals into frequency-domain signals. The transformation condenses an infinite number of uniformly distributed signal values into a limited set of frequency-based components, enabling the system to detect attacks associated with various forms of traffic behavior. These collected features are then used for intrusion detection classification, which aids in the accurate detection of unusual or malicious behavior. Eq. (2) represents the mathematical transform of N points. Where $\omega_N = e^{-\frac{2\pi i}{N}}$ is an Nth root of the unit, j or $x(j)$ is a sample index in the time domain series, and $X(k)$ is a frequency-domain component at index k.

$$X(k) = \sum_{j=1}^{N} x(j)\omega_N^{(j-1)(k-1)} \tag{2}$$

The generated IoT sensor data used in the intrusion detection assessment was discrete-time changing. Trimmed data was transformed into a frequency domain using the FFT tool. The prior approach yielded a series of arbitrary intensities ($\{a_i\}$), which were subsequently converted to an intrusion performance on their frequency ($\{|a_i|\}$). The scaling process was required as the level of magnitude of a material ($\{|a_i|\}$) fluctuated, as shown in Eq. (3).

$$\{|a_i|\} = 10log\{a_i\} \tag{3}$$

Threat detection using gradient boosting based adaptive artificial gorilla troops (GraBoost-AAGT)

The proposed GraBoost-AAGT is a hybrid intrusion detection approach that combines the strengths of AAGT and GraBoost for ultra-low power threat detection in smart and green IoT systems. AAGT, inspired by the collective intelligence of gorilla troops, is used to optimize feature selection, reducing dimensionality and computational overhead to suit resource-constrained devices. High detection accuracy and model adaptability to changing threats are guaranteed by GraBoost's adaptive learning capabilities. The hybrid integration enables lightweight deployment with minimal memory and CPU usage, making it suitable for TinyML environments. This synergy ensures efficient, real-time threat detection while supporting energy-efficient operation critical for sustainable IoT infrastructures like smart homes, cities, and agricultural systems.

Gradient boosting (GraBoost)

The proposed GraBoost leverages the advanced supervised learning capabilities of GraBoost within the Gradient Boosting framework to effectively detect network intrusions. To enhance detection accuracy and prevent overfitting, the model integrates a regularization term into the objective function, which smooths the learned weights. It further improves optimization by incorporating both first- and second-order gradient statistics during loss function minimization. For robust intrusion detection, the model adopts row and column sampling, which enhances generalization and reduces variance. Additionally, parallel and distributed computing mechanisms are employed to accelerate model training, enabling quicker exploration of potential threats. These collective steps, including regularization, gradient optimization, sampling strategies, and computational efficiency, make GraBoost a powerful and adaptive solution for intrusion detection tasks. In the paragraphs that follow, the GraBoost framework is briefly explained. Eq. (4) states that the gradient boosting tree model's estimated output $\widehat{z}_j$ is equal to the sum of the prediction scores, $e_l(Y_j)$ for each tree.

$$\widehat{z}_j = \sum_{l}^{L} e_l(Y_j), e_l \in \Gamma \tag{4}$$

Where each $e_l \in \Gamma$ and Γ represents the space of regression trees. Here, L denotes the total number of trees, and Y_j corresponds to the feature set of samples j. For each leaf node j, a prediction score $e_l(Y_j)$ is known as the protocol that is allocated, representing the regression value for all samples falling into that node. The leaf weight

$\omega_i$corresponds to node $i \in \{1, 2, \ldots, T\}$, where T represents all of the tree's leaf nodes. These predictive scores across trees are aggregated to classify traffic behavior and identify potential intrusions in the network.

The regularized objective function in the intrusion detection model combines a loss function as well as a regularization term, allowing each model to accurately learn patterns from training data while preventing overfitting. Every loss function procedure calculates the prediction error, ensuring the model detects intrusions effectively, while the regularization term controls model complexity by penalizing overly complex trees, thus improving generalization to unseen threats.

$$\Phi = \sum_{i=1}^{n} m(z_j, \widehat{z}_j) + \gamma T + \frac{1}{2}\lambda \sum_{i-1}^{T} \omega_i^2 \tag{5}$$

Let Eq. (5), n and Φ represent the quantity of training samples with overall objective function value, and $m(z_j, \widehat{z}_j)$ are loss functions measuring the prediction error between actual output z_j and predicted output $\widehat{z}_j$. The term γT penalizes the total number of leaf nodes T, where γ is the complexity cost per leaf. The final term $\frac{1}{2}\lambda \sum_{i-1}^{T} \omega_i^2$ applies L2 regularization to the leaf weights ω_i^2, with λ controlling the degree of regularization to prevent overfitting in intrusion detection.

The additive learning method builds trees individually, with each new tree learning from its predecessors and updating prediction values. Where each new tree $e_l\left(y_j\right)$ corrects the residuals from the previous prediction $\widehat{z}_j^{(l-1)}$, and $\Phi_{(l)}$ are the objective function at iteration. Thus, the updated prediction at the lth iteration becomes $\widehat{z}_j^{(l)} = \widehat{z}_j^{(l-1)} + e_l\left(y_j\right)$. The corresponding regularized function at iteration l is defined as Eq. (6).

$$\Phi_{(l)} = \sum_{j=1}^{n} m\left(z_j,\ \widehat{z}_j^{(l-1)} + e_l\left(y_j\right)\right) + \frac{1}{2}\lambda \sum_{i-1}^{T} \omega_i^2 \tag{6}$$

The first-term loss function is approximated by a second-order Taylor expansion to increase efficiency in maximizing the goal. By capturing both the gradient and curvature information, this approximation allows for quicker and more accurate model updates. Consequently, it improves threat detection accuracy and speeds up convergence by more effectively directing the optimization process in every training cycle, as shown in Eq. (7).

$$\Phi_{(l)} \cong \sum_{j=1}^{m} \left[k\left(z_j, \widehat{z}_j^{(l-1)}\right) + h_j e_l(w_j) + \frac{1}{2} g_j e_l^2(w_j)\right] + \gamma T + \frac{1}{2}\lambda \sum_{i=1}^{S} \omega_i^2 \tag{7}$$

Here, $h_j = \partial_{\hat{z}^{(l-1)}} k\left(z_j, \widehat{z}^{(l-1)}\right)$ and $g_j = \partial^2_{\hat{z}^{(l-1)}} k\left(z_j, \widehat{z}^{(l-1)}\right)$ represent the loss function's initial and another order gradient data. Removing constant terms yields the approximate objective in step l, as follows in Eq. (8).

$$\Phi_{(l)} \cong \sum_{j=1}^{m} \left[h_j e_l(w_j) + \frac{1}{2} g_j e_l^2(w_j)\right] + \gamma T + \frac{1}{2}\lambda \sum_{i=1}^{S} \omega_i^2 \tag{8}$$

A tree is defined by a vector of scores allocated to its leaf nodes, and an element index mapping function that assigns each instance to a certain leaf g_j. This process is denoted as $\sum_{i=1}^{n} f_k = \sum_{j=1}^{T} \omega_i$, and Eq. (9) can be revised to reflect this structure.

$$\Phi_{(l)} = \sum_{i=1}^{S} \left[\left(\sum_{j\in J_i} h_j\right)\omega_i + \frac{1}{2}\left(\sum_{j\in J_i} g_j + \lambda\right)\omega_i^2\right] + \gamma T \tag{9}$$

In intrusion detection, given a specified tree structure, the optimal leaf weights ω_i^* at each node are determined by minimizing the regularized objective function $\Phi_{(l)}^*$. The optimization is carried out via quadratic programming, which allows for the effective computation of weight values that improve the performance accuracy in characteristic between normal and malicious traffic patterns, as follows in Eqs. (10–11).

$$\omega_i^* = \frac{\sum_{j\in J_i} h_j}{\sum_{j\in J_i} g_j + \lambda} \tag{10}$$

$$\Phi_{(l)}^* = -\frac{1}{2}\sum_{i=1}^{S} \frac{\left(\sum_{j\in J_i} h_j\right)^2}{\sum_{j\in J_i} g_j + \lambda} + \gamma T \tag{11}$$

Eq. (11) functions as a structure scoring function, determining how well a particular vector of leaf scores fits the data. Lower numbers suggest a better fit and more efficient model performance. Because there are infinitely many alternative tree structures, a greedy approach is commonly employed to efficiently discover the best tree structure for intrusion detection.

Adaptive artificial gorilla troops (AAGT)

The Gorilla Troops Optimizer (GTO) is a swarm-based metaheuristic algorithm that draws stimulation from intrusion detection in IoT devices. A normal gorilla tribe is led by a dominant silverback male and includes numerous females and offspring. The algorithm is modeled after gorilla behaviors such as group formation, territory defense, and male dominance and mating rivalry. AAGT comprises two primary phases: exploration and exploitation. To search the solution space during the exploration phase, it employs three strategies: going to an undiscovered site, moving near other gorillas, and returning to a known position. During the exploitation phase, two operators watch the silverback and compete for females to refine solutions and improve convergence to ideal results.

Three different operators, one of data, visited a new site, which was used to further the investigation during the exploratory phase. The second component improves the balance between exploration and exploitation by switching to various attacks in IoT network traffic. The AAGT algorithm's capacity to explore different emerging regions is significantly improved by shifting to a familiar place during the exploration phase, which is the third component. Eq. (12) provides a symbolic representation of these discrete operators.

$$HW(s+1) = \begin{cases} (ub - lb) \times q_1 + lb \; rand < y \\ (q_2 - D) \times W_q(s) + C \times A \; rand \geq 0.5 \\ W(j) - C \times \left(C \times \left(W(s) - HW_q(s)\right) + q_3 \times \left(W(s) - HW_q(s)\right)\right) rand < 0.5 \end{cases} \tag{12}$$

Where, $D = (Cos(2 \times q_4) + 1) + \left(1 - \frac{it}{Maxit}\right)$ represents a dynamic factor influenced by a cosine function and a time decay term $\left(1 - \frac{it}{Maxit}\right)$, which helps adjust the search intensity over iterations. $C = D \times l$ scales this factor D by a random variable l, introducing stochastic

Algorithm 1
. GraBoost-AAGT.

Input: Objective function f(x), population size N, max iterations T, dimension d, initial parameters, and optional GraBoost model.

Output: *Best solution found x_best and its fitness f_best*
Step 1: *Gradient Boosting Assistance:*
Option A: *Use GB to predict f(x_i new) and rank candidates*
Option B: *Update GB model incrementally with (x_i new, f_new)*
Option C: *Use GB to refine poor solutions (e.g., local adjustment)*
Accept x_i new if f_new is better than f(x_i)
Update x_i and f(x_i) accordingly
Step 2: *Update the silverback (leader) of x_i new*
Optional: *Maintain population diversity (e.g., reinitialization of stagnant gorillas)*
Start 3: *Initialize: Generate initial population of gorillas: X = {x₁, x₂, …, x_N}*
- Evaluate fitness: F = {f(x₁), f(x₂), …, f(x_N)}
- Identify leader (silverback): x_best = argmin f(x), f_best = f(x_best)
Start 4. *For t = 1 to T do: Adaptive Parameter Update:*
Update control parameters (e.g., step size α(t), exploration rate β(t))
Possibly adjust based on convergence curve or diversity metrics
Step 5: *For each gorilla x_i in X do:*
- Generate random number r ∈ [0, 1]
If r < β(t):
- Apply exploration rules (e.g., migration, social movement)
- x_i new = ExplorationMove(x_i, α(t), random_neighbor)
Else:
- Apply exploitation rules (e.g., competing for females)
- x_i new = ExploitationMove(x_i, x_best, α(t))
- Evaluate new fitness: f_new = f(x_i new)
Step 6. *Return x_best and f_best*
End

variation to guide search diversity. $F = [-D, D]$ is a vector within the range $[-D, D]$, determining the direction and magnitude of the intrusion detection in the search space. Finally, $A = F \times W(s)$ is calculated as the product of F and a weighting function $W(s)$, which modulates the influence of the search step based on current conditions.

Where the location of the gorilla candidate in the subsequent iteration is indicated by $HW(s+1)$. The variables' lower and upper bounds are shown by lb and ub. Random values between 0 and 1 are denoted by the symbols q_1, $rand$, q_2, q_3, and q_4. The range of the parameter y is 0 to 1. The gorilla position's current vector is represented by $W(s)$, and a randomly chosen gorilla member is indicated by $W_q(s)$ and $HW_q(s)$. A random value between −1 and 1 is indicated by the symbol l.

There are two actions or strategies used in the exploitation phase. When a specific criterion $C \geq W$ is satisfied, the first tactic, referred to as "Follow the silverback," is used. In this case, W stands for a preset parameter that was established before the optimization procedure. The first tactic can be calculated mathematically using Eq. (13).

$$HW(s+1) = C \times N \times (W(s) - W_{silverback}) + W(s)$$

$$N = \left(\left| \frac{1}{M} \sum_{j=1}^{M} HW_j(s+1) \right|^h \right)^{\frac{1}{2}}$$

$$h = 2^C \tag{13}$$

Where N is the overall amount of attack behavior, $W_{silverback}$ is the best response, and $HW(s+1)$is the threat's updated position in the examination space at the subsequent iteration. By integrating this technique into the optimization procedure, the algorithm explores potential sections of the search space using data from the greatest result that has been found at the moment in intrusion detection. Eq. (14) is essential for directing the search space's exploration and exploitation, which affects the algorithm's capacity to identify the best solutions. Where β denotes a parameter that needs to be set before the optimization process. A random number between 0 and 1 is indicated by the character q_5.

$$HW(j) = W_{silverback} - (W_{silverback} \times R - W(s) \times R) \times B$$

$$R = 2 \times q_{5-1}$$

$$B = \beta \times G$$

$$G = \begin{cases} M_1, rand \geq 0.5 \\ M_2, rand < 0.5 \end{cases} \tag{14}$$

The technique was created to mitigate the possibility that the optimal value might become trapped in local optima. The controller's parameters include fractional order control and proportional-integral-derivative (PID) techniques, among other optimization concerns. Based on this approach, several contemporary solutions have been developed and used in a variety of optimization scenarios in intrusion detection. This progress is determined by two phases. The high-velocity ratio scenario is the initial stage. Eqs. (15–17) represent this stage's mathematical model.

$$it < \frac{1}{3} Maxit \tag{15}$$

$$T = \vec{Q_A} \otimes \left(F - \vec{Q_A} \otimes W_j(s) \right) \tag{16}$$

$$W_j(s+1) = W_j(s) + O.\ \vec{Q_A}(T) \tag{17}$$

Where $\vec{Q_A}$ denotes the normal distribution's Brownian motion-reflecting vector of random integers. Entry-by-entry multiplications are denoted by the symbol $\otimes$. In the interval [0,1], $\vec{Q_A}$ is a sequence of uniform random values, and $P = 0.5$ is a constant. The previous position is multiplied by $\vec{Q_A}$ to create the new location. This state, which is indicative of a high degree of exploratory capacity, occurs throughout the first third of variations when the number of steps is large. Eq. (18) determines that the optimum place to create a matrix is the fittest solution (F). To generate the F matrix, the superlative result, Wa, is differentiated m times. c represents the number of dimensions, while m indicates the number of look agents.

$$F = \begin{bmatrix} Wa^s_{1,1} & \cdots & Wa^s_{1,c} \\ \vdots & \ddots & \vdots \\ Wa^s_{m,1} & \cdots & Wa^s_{m,c} \end{bmatrix} \tag{18}$$

Table 1
Overall system configuration.

Parameter	Value
Programming language	Python 3.6.13
CPU	Intel Core i5 10600KF
OS	Windows 11 Pro
RAM	32 GB
GPU	NVIDIA RTX2060

The low-velocity ratio is the second stage. This phase, which is usually associated with strong exploitation capability, occurs close to the conclusion of the optimization process. Position updating is aided by adding the process size to the location and expanding F in the Lévy procedure. Increasing the likelihood of escaping local minima is an additional characteristic of AAGT.

The GraBoost-AAGT hybrid model combines GraBoost's robust classification performance with AAGT's adaptive exploration–exploitation balance. GraBoost guarantees high-accuracy detection through stage-wise learning and robust error correction, while AAGT improves the model's capacity to dynamically seek complex solution spaces by optimizing feature selection and parameter adjustment for threat patterns. The synergy is perfect for edge-based security systems because it allows for extremely low power consumption, real-time adaptation, and high detection precision in energy-constrained contexts. Algorithm 1 shows the proposed GraBoost-AAGT method.

Result and discussion

The section describes the hardware, datasets, and operation environment used in the experiment. It displays evaluation parameters including specificity, computation time, recall, accuracy, and precision. After analyzing the overall performance results, the suggested method is evaluated in comparison to other methods. Ultimately, the results are interpreted through a thorough discussion that highlights advancements, constraints, and practical deployment implications.

Experimental setup

The system comprises modular libraries for data acquisition, instrument control, and automation of laboratory experiments, supporting diverse lab devices through user-friendly interfaces. The GraBoost-AAGT model is integrated for intelligent decision-making, optimizing experimental parameters with ultra-low power consumption and memory-efficient operations. Overall performance evaluation configuration is illustrated in Table 1.

Evaluation metrics

The algorithms were fairly evaluated using the metrics listed below. True positive (TP): The number of malicious traffic incidents is accurately detected as intrusion attack. True negative (TN): The number of traffic incidents that were accurately perceived as non-threat. False positive (FP): The size of benign traffic cases is incorrectly stopped as harmful. False negative (FN): The quantity of malicious traffic cases that were mistakenly classified as benign.

Accuracy: It is the percentage of accurately detected incidents, both real attacks and everyday activities, among all cases that were assessed. It provides a broad performance indicator that fails to consider dataset class imbalance. It shows how well the model performs generally in differentiating between safe and dangerous behavior, as shown in Eq. (22).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{22}$$

Precision: It is the percentage of threats that were correctly identified out of all the instances that the system detected as threats. High precision means fewer false alarms, ensuring the system alerts only for actual malicious activities, which is crucial for reducing unnecessary responses and conserving resources in IoT security, as shown in Eq. (23).

$$Precision = \frac{TP}{TP + FP} \tag{23}$$

Recall/Sensitivity: The model's capacity to accurately distinguish every true threat from the entire number of threats that exist. A high recall ensures that the majority of harmful activity is recognized, even in the event of some false alarms, because fewer threats are overlooked (Eq. (24)).

$$Recall = \frac{TP}{TP + FN} \tag{24}$$

F1-score: It is a statistic that calculates the harmonic mean of intrusion detection for precision and recall scores. Since it considers both FP and FN in the same computation, it is regarded as a continuation of the accuracy measure. Eq. (25) is utilized for its computation.

$$F_1 = 2 * \frac{Precision * Recall}{Precision + Recall} \tag{25}$$

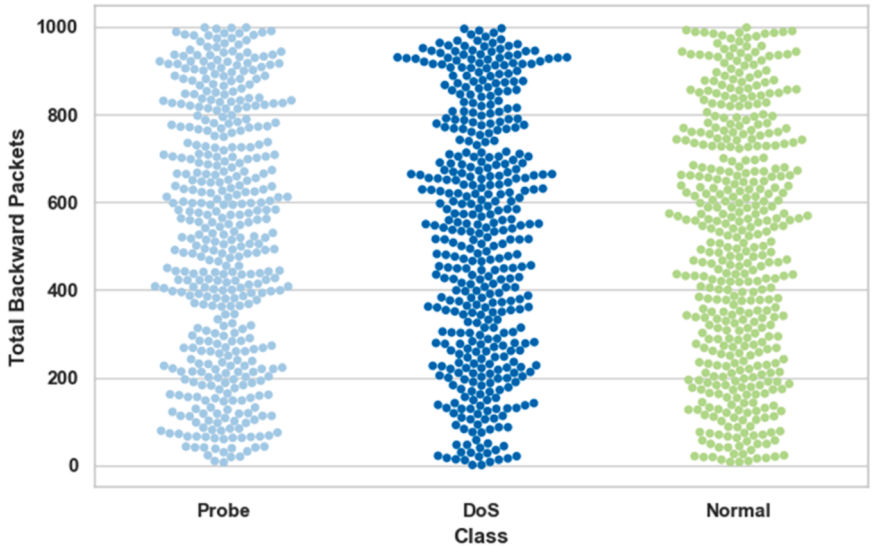


Fig. 2. Outcome of swarm: backward packets by class.

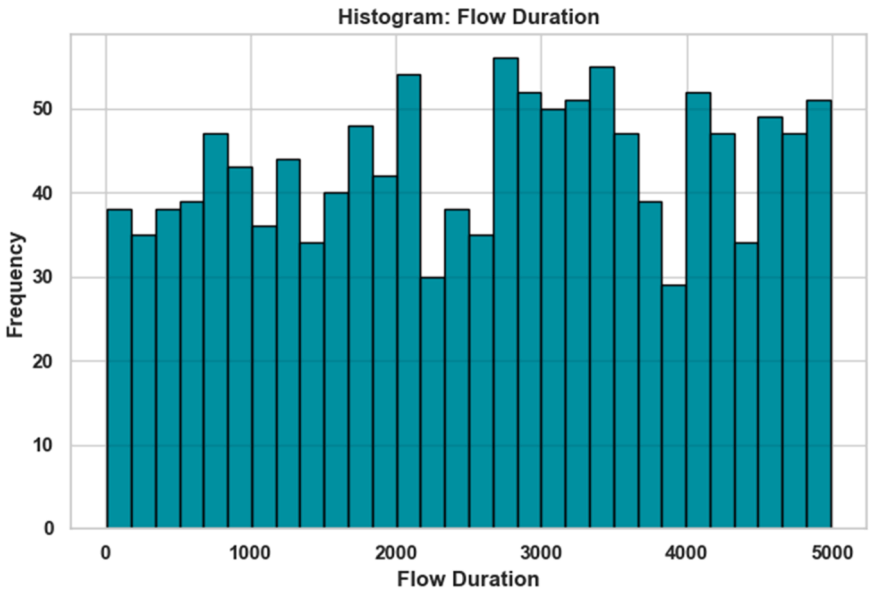


Fig. 3. Histogram: flow duration.

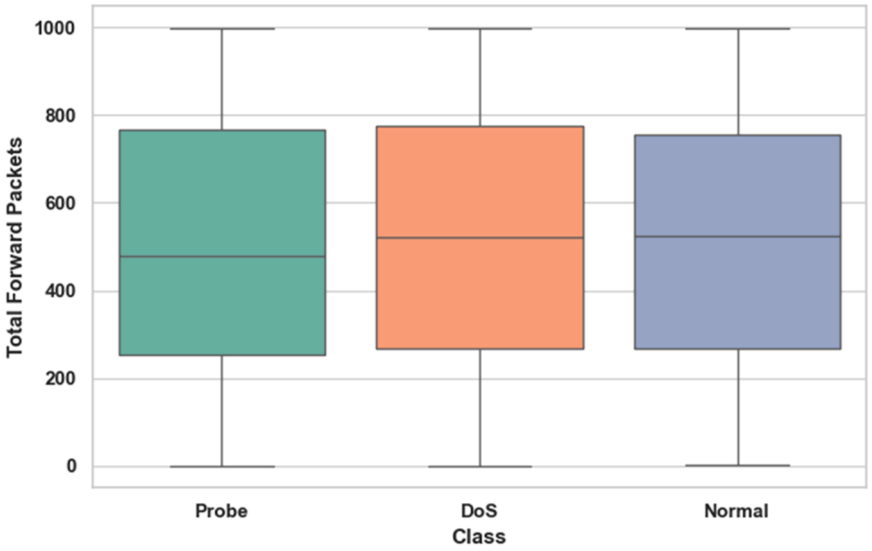


Fig. 4. Boxplot: forward packets by class.

Specificity: It measures the ability of an intrusion detection system to correctly identify non-malicious traffic. It is the proportion of true negatives (correctly detected benign traffic) out of every actual non-malicious instance, indicating how well the system avoids false alarms.

Computation time: It refers to the total duration an intrusion detection system takes to process data and produce detection results. Lower computation time is crucial for real-time detection in energy-constrained IoT devices, ensuring timely response without excessive resource consumption.

Performance outcome

The distribution of all backward packets among the three classes, Probe, DoS, and Normal, is shown in Fig. 2. Packet counts for all classes range from practically 0 to over 1000, with DoS showing the densest clustering at higher levels, signifying patterns of intense traffic. In comparison to DoS, Probe and Normal have somewhat lower packet densities in the higher range, but their distributions are similar. This pattern implies that while Probe and Normal maintain more balanced, modest traffic levels, DoS attacks consistently produce significant backward packet quantities.

Fig. 3 shows the histogram of flow duration *versus* frequency. The flow durations range from nearly 0 to 5000 units, with frequency values mostly between 30 and 55. The distribution appears relatively uniform, without extreme peaks or drops, indicating balanced occurrences of flow durations. Higher frequencies above 50 occur intermittently, suggesting occasional bursts in network activity.

Fig. 4 illustrates the distribution of total forward packets for Probe, DoS, and Normal classes. DoS and Normal classes exhibit slightly higher median forward packet activity compared to Probe. All classes show a wide range, indicating variability in packet transmission patterns across different traffic types.

The number of distinct protocol types is shown in Fig. 5. The highest count is displayed via user datagram protocol (UDP), which indicates larger network utilization and, in some cases, possibly superior performance. With a somewhat lower count, internet control message protocol

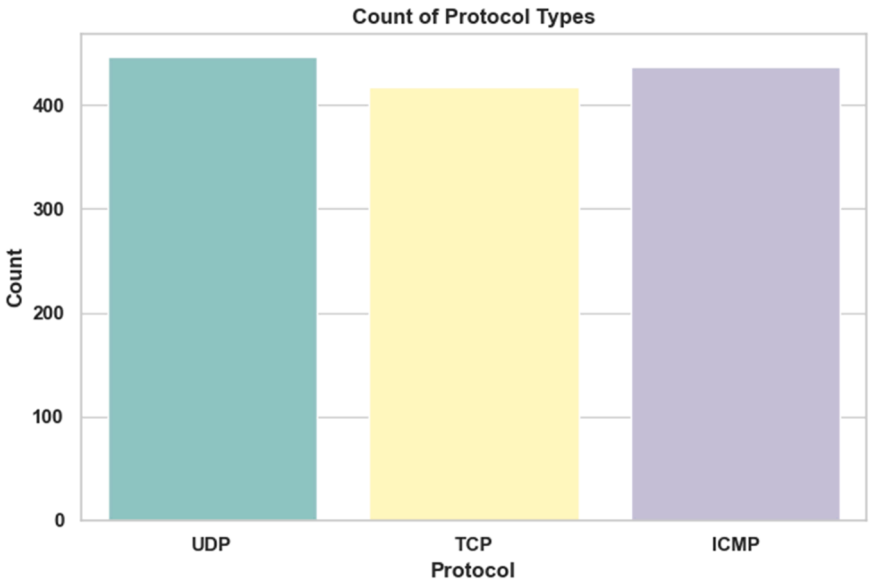


Fig. 5. Count of protocol types.

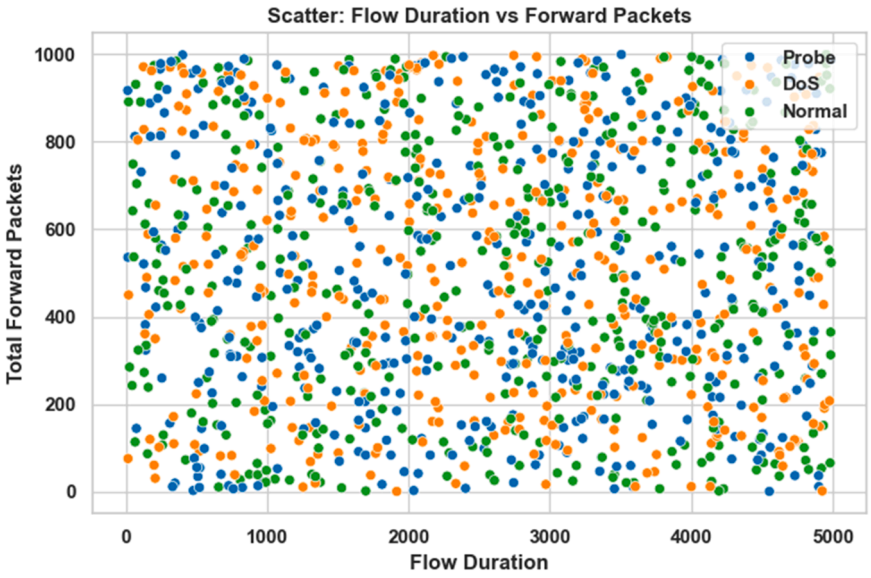


Fig. 6. Scatter for duration *vs* Forward packets.

(ICMP) comes next, signifying moderate activity. Compared to UDP and ICMP, transmission control protocol (TCP) has the lowest count, indicating a relatively lesser usage and performance frequency in the examined dataset.

Fig. 6 shows the relationship between flow duration and total forward packets for Probe, DoS, and Normal traffic. Higher flow durations often correspond to a wide range of packet counts, from low to high. DoS attacks frequently appear across all ranges, while Probe and Normal traffic also show varied packet volumes, indicating diverse network behaviors.

Fig. 7 shows the distribution of the mean Inter-Arrival Time (IAT) across three classes: Probe, DoS, and Normal. The violin plot reveals similar median IAT values for all classes, with wide variability. No class distinctly stands out with a higher or lower mean IAT, indicating moderate performance differentiation.

Fig. 8 shows a Kernel Density Estimation (KDE) plot of average packet sizes in a network dataset. It illustrates the probability density distribution, with most packet sizes ranging. Peaks indicate common sizes, while the smooth curve reveals the overall distribution pattern, aiding in network traffic analysis and anomaly detection.

Fig. 9 shows the strip plot that visualizes the distribution of Flow Bytes/s across three traffic classes: Probe, DoS, and Normal. Each dot represents an individual flow measurement, with values ranging from near zero to approximately bytes per second. The clustering patterns highlight variations in flow volume across categories, enabling comparison of traffic behavior and identification of potential anomalies in network performance or attack patterns.

Comparison phase

The comparative analysis evaluates the proposed GraBoost-AAGT method against existing methods such as Random Forest [26], Adaptive Boosting (AdaBoost) [26], and Firefy Algorithm-ML (FA-ML) ("FA-ML approach uses an SVM for classification, and the Grey Wolf Optimizer (GWO) algorithm is used to tune the parameters") [27] for intrusion detection, highlighting its improved feature extraction,



Fig. 7. Violin: IAT mean by class.

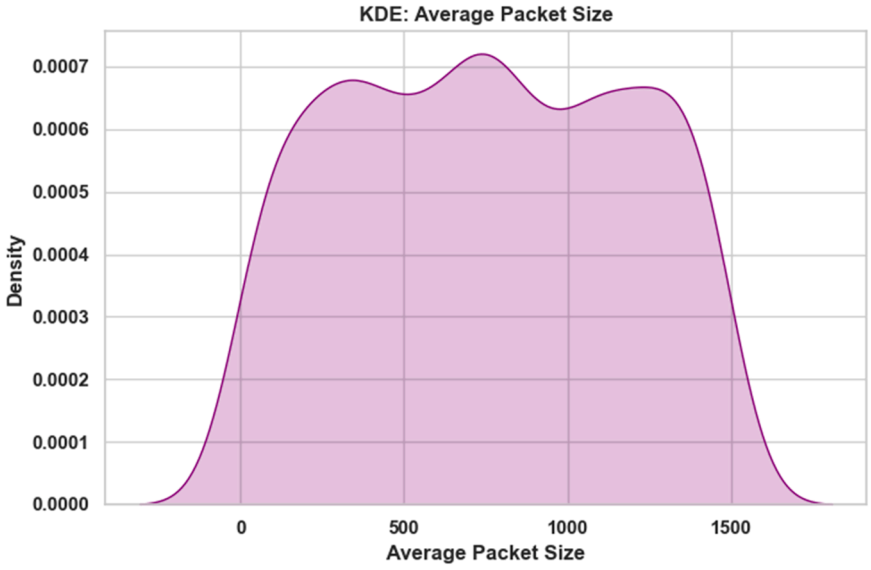


Fig. 8. KDE: Average packet size.

classification effectiveness, and robustness, resulting in enhanced detection accuracy and reliability over existing approaches. Table 2 shows the overall comparison performance values for the existing and proposed methods.

As demonstrated in Table 2 and Fig. 10, accuracy is defined as the percentage of correctly identified cases across all samples. The proposed GraBoost-AAGT method achieved 99.50 %, superior Random Forest (99.39 %), AdaBoost (95.26 %), and FA-ML (99.34 %). It is extremely dependable for real-world intrusion detection in IoT contexts due to its exceptional classification capacity, which successfully reduces false positives and false negatives.

Table 2 and Fig. 11 show the performance evaluation for the existing and suggested methods in precision. GraBoost-AAGT achieved 99.8 %, which is slightly better than Random Forest (99.7 %) and significantly higher than AdaBoost (95.7 %). The high precision of GraBoost-AAGT reflects its ability to avoid false alarms by ensuring most alerts correspond to actual threats.

Table 2 and Fig. 12 show that recall measures the proportion of actual threats correctly detected. The proposed GraBoost-AAGT method achieved a higher recall of 99.12 % compared to Random Forest (98.9 %), FA-ML (98.36 %), and AdaBoost (94.5 %). This superior performance demonstrates its effectiveness in detecting nearly all intrusions, thereby minimizing undetected attacks and enhancing IoT network security.

Fig. 13 and Table 2 illustrate the F1-score performance across different methods. The proposed GraBoost-AAGT achieved the highest F1-score of 99.45 %, slightly surpassing Random Forest (99.4 %), and significantly outperforming FA-ML (96.67 %) and AdaBoost (95.1 %). This high F1-score reflects the proposed method's strong balance between precision and recall, ensuring accurate and consistent intrusion detection in IoT environments.

Table 2 and Fig. 14 show that specificity evaluates the proportion of actual safe instances correctly identified. GraBoost-AAGT scored 99.81 %, surpassing FA-ML (99.65 %). High specificity ensures that the method rarely misclassifies benign traffic as malicious, reducing false positives and preserving efficiency in energy-constrained IoT systems where unnecessary detections waste critical resources.

Fig. 15 and Table 2 show the computation time comparison, where

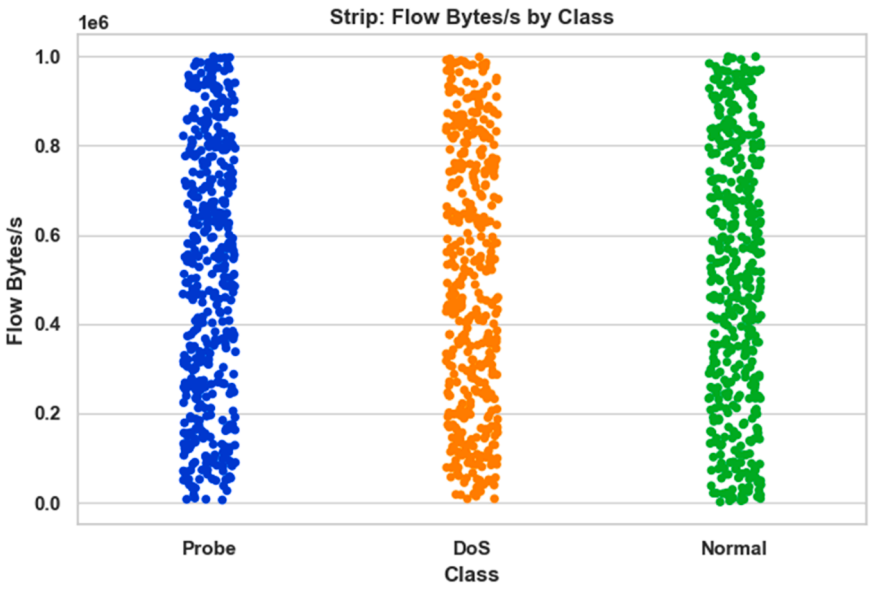


Fig. 9. Strip: flow bytes/s by class.

Table 2
Numerical outcomes of existing and suggested methods.

Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Specificity (%)	Computation time (*sec*)
Random Forest [26]	99.39	99.7	98.9	99.4	–	–
AdaBoost [26]	95.26	95.7	94.5	95.1	–	–
FA-ML [27]	99.34	–	98.36	96.67	99.65	6.3
GraBoost-AAGT [Proposed]	99.50	99.8	99.12	99.45	99.81	4.5

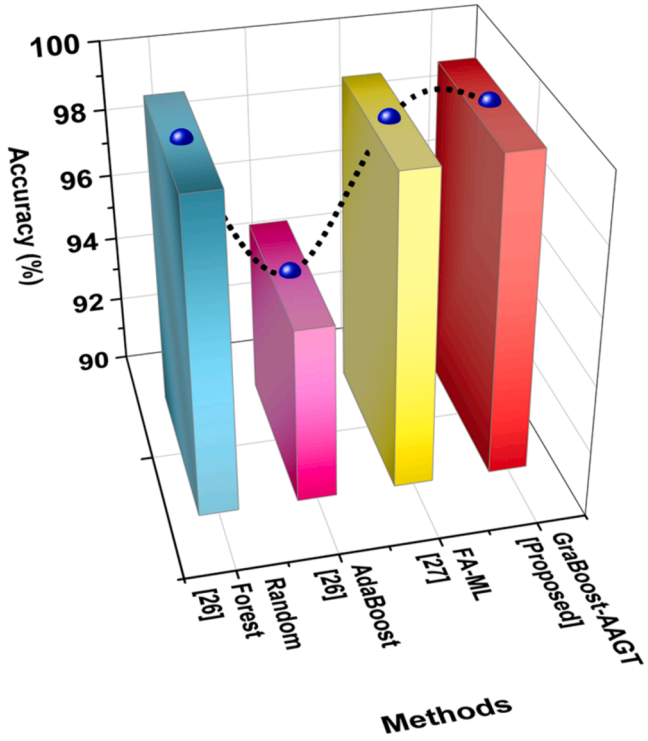


Fig. 10. Performance evaluation of the accuracy.

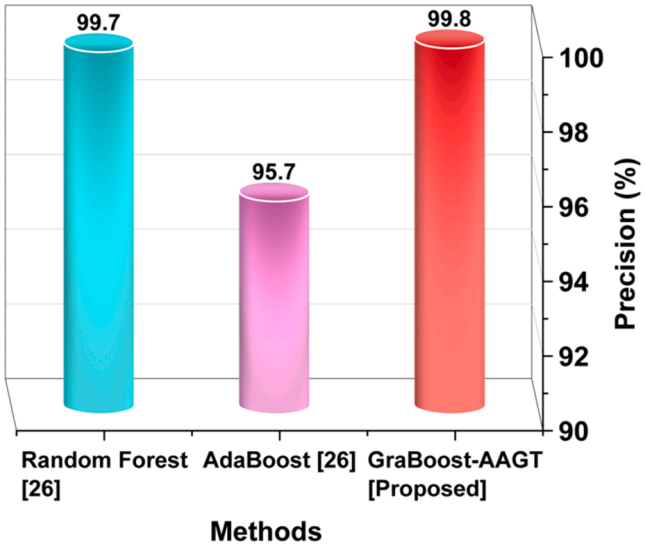


Fig. 11. Comparison of the performance of the precision.

the proposed GraBoost-AAGT achieved the lowest time of 4.5 s, outperforming FA-ML (6.3 s) and other methods with unavailable values. This lower computation time highlights the proposed method's efficiency, enabling faster threat detection and reduced resource usage in IoT environments.

Discussion

In the realm of sustainable and energy-constrained IoT devices, IDS must balance high detection accuracy with minimal resource consumption. Traditional ML methods like Random Forest [26], AdaBoost [26], as well as FA-ML [27], have been widely used for IDS. However, these approaches often suffer from limitations such as insufficient

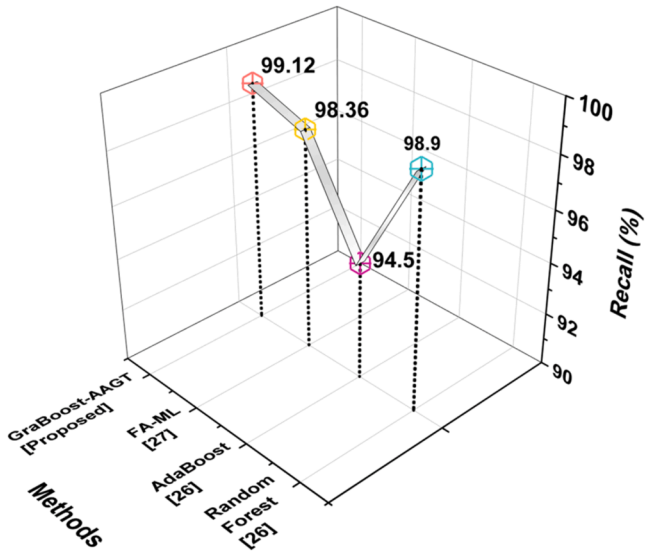


Fig. 12. Performance evaluation of the recall.

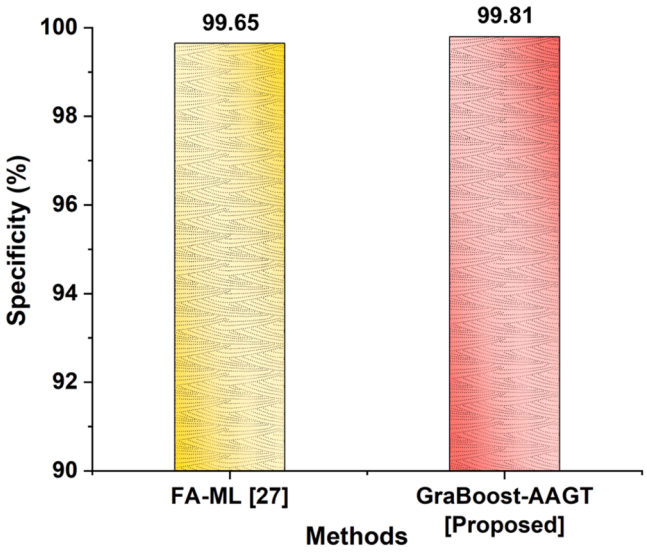


Fig. 14. Overall evaluation of the specificity.

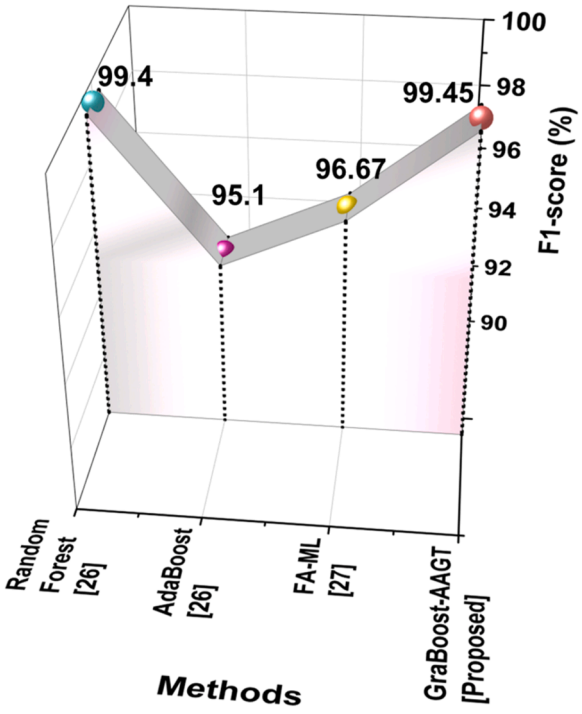


Fig. 13. Evaluation of the existing and proposed method in F1-score.

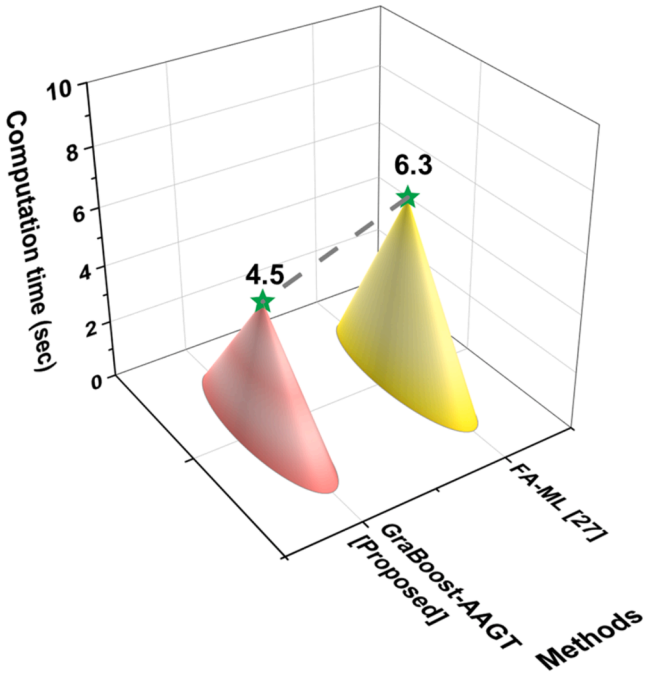


Fig. 15. Performance evaluation of the computation time.

feature extraction capabilities, susceptibility to overfitting, and suboptimal classification performance in resource-constrained environments. Random Forest and AdaBoost, while effective in certain contexts, can be computationally intensive and may not fully capture complex intrusion patterns. The FA-ML method, despite its optimization focus, sometimes lacks robustness against diverse attack types. To address these challenges, the proposed GraBoost-AAGT method integrates gradient boosting with an adaptive attention-guided transformer, specifically designed for TinyML environments. The method improves feature extraction by concentrating on the most pertinent data patterns, which increases system robustness and classification efficacy. Comparative analysis demonstrates that GraBoost-AAGT outperforms Random Forest, AdaBoost, and FA-ML methods, offering superior intrusion detection accuracy and reliability while maintaining low energy consumption, making it highly suitable for sustainable IoT device security.

Conclusion

In energy-constrained and ecologically sensitive situations, the necessity for effective, real-time intrusion detection has increased due to the rapid deployment of IoT devices. Traditional IDS solutions are not appropriate for widespread IoT deployment, as their solutions frequently fall short of the low-power and low-memory demands of embedded systems. To address this challenge, this research proposed the GraBoost-AAGT method, an integration of AAGT optimization and XGBoost optimized for ultra-low power consumption and memory efficiency. It incorporates categorical encoding, Z-score normalization, and FFT-based feature extraction to ensure lightweight and accurate threat detection. Extensive evaluations using diverse IoT datasets across cloud,

edge, and TinyML-based deployments show that the TinyML implementation achieves superior detection performance, with over 99.50 % accuracy, 99.8 % precision, 99.12 % recall, 99.45 % F1-score, 99.81 % specificity, and computation time (4.5 s) while maintaining minimal memory and energy usage. Despite its promising results, limitations include initial deployment costs, energy constraints under high-frequency attacks, and restricted data transfer capabilities in ultra-low power environments. Future work will focus on enhancing model compression techniques, integrating ML methods to reduce data transmission, and expanding detection capabilities to support a broader range of IoT protocols and threats. The approach lays the foundation for secure, scalable, and sustainable IoT security architectures.

CRediT authorship contribution statement

Amuthadevi C: Formal analysis, Data curation, Conceptualization. **Venkatesan R:** Writing – review & editing, Writing – original draft, Validation. **Mythily M:** Resources, Methodology, Investigation. **Aroul Canessane R:** Writing – review & editing, Resources, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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